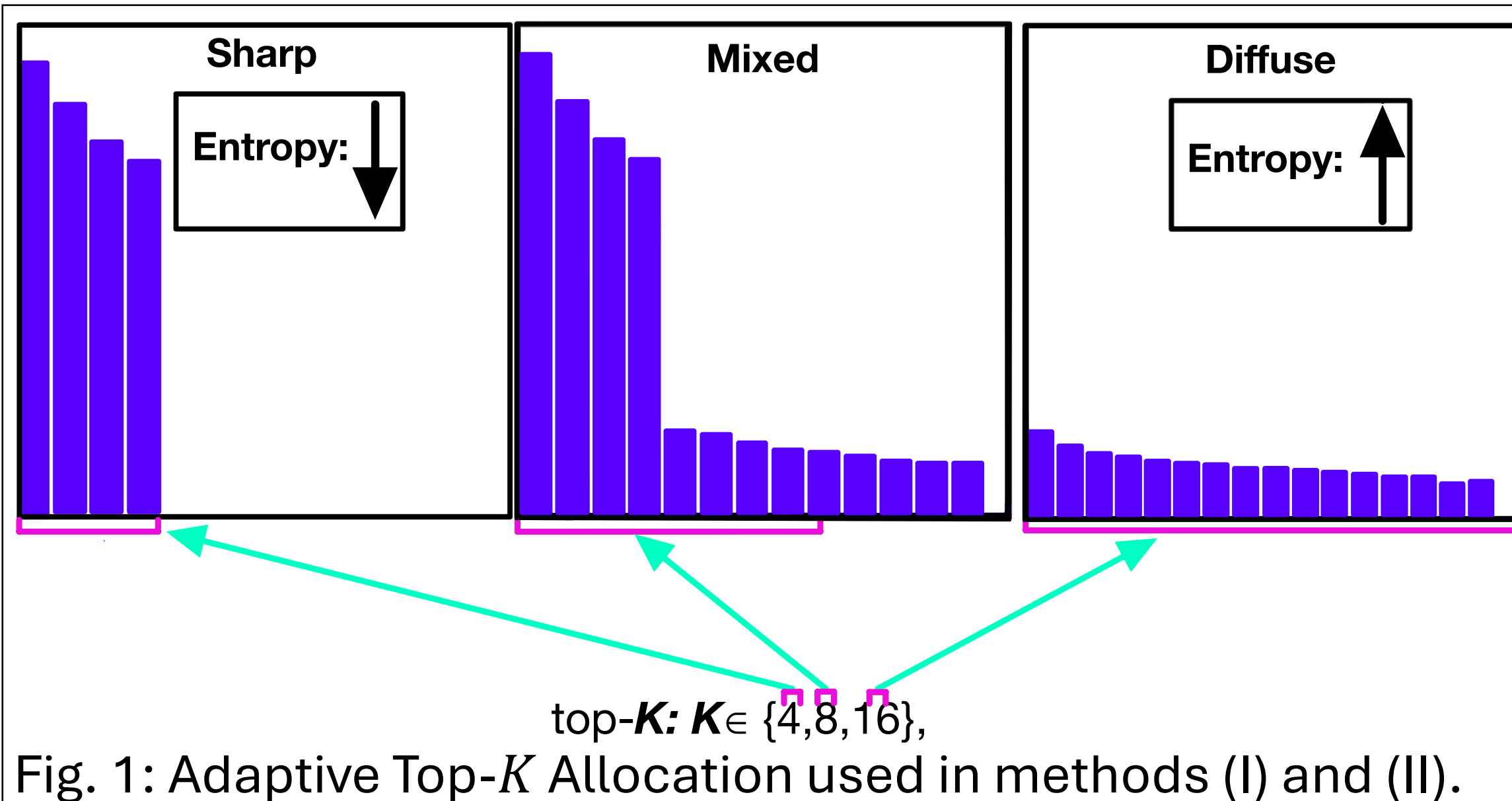


When Full KD Falls Short: Adaptive Sparse Distillation for Small Language Models

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INTRODUCTION

- In Knowledge distillation (KD) for autoregressive language modeling, the teacher’s next-token distribution provides richer supervision than hard labels.
- Storing or recomputing the full distribution over a large vocabulary at every token position is expensive.
- Sparse methods, such as Top- K truncation and sampling-based approximations, lower this storage overhead.^{1,2}
- Most sparse KD work views the full teacher distribution as the ideal target and sparsification as lossy compression.
- We question this assumption for large-teacher/small-student distillation.
- Prior work and our baseline experiments suggest that sparse KD may match or outperform full KD.^{1,2}
- For a large teacher/small student setup, full KD may fall short in high-uncertainty contexts, because matching a diffuse tail may dilute training gradients and consume student capacity, distracting from the most useful structure in the teacher’s predictions.
- We reframe sparse KD as both an efficiency and supervision-filtering solution, proposing two adaptive sparse methods that outperform full KD in our setup.

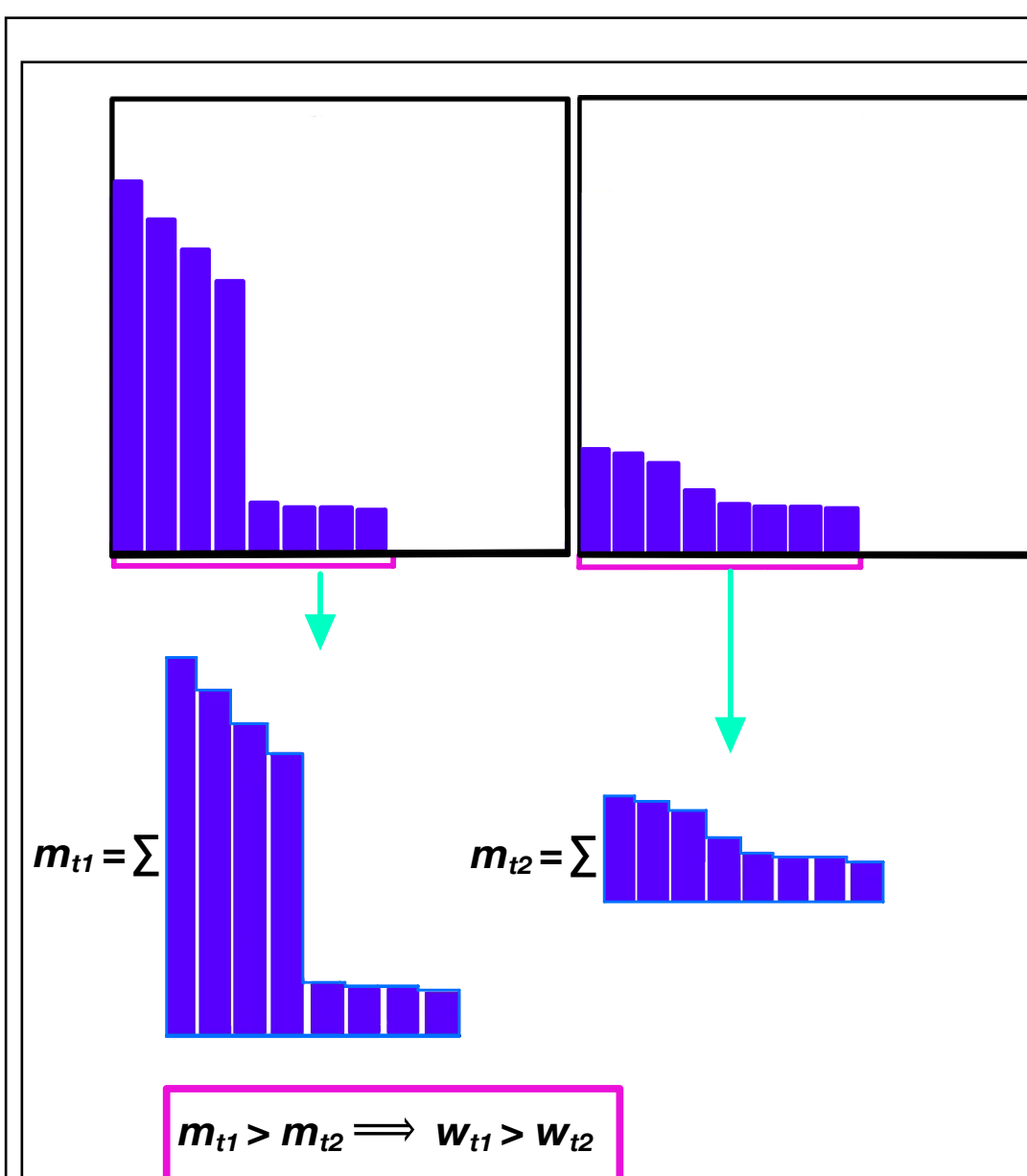


Fig. 2: Retained-head-mass weighting used in method (II).

EXPERIMENTAL DESIGN

- Teacher:** Pythia-1.4B
- Student:** Pythia-160M
- Dataset:** WikiText-103
- Sequence length:** 256
- Train examples:** 200,000
- Baseline methods:**
 - Full KD
 - Vanilla Top- K
 - Sampling-based Approx.
- Metrics:**
 - Negative-Log-Likelihood
 - Perplexity
 - Stored scalars per token

PROPOSED METHODS

(I) Adaptive Top- K Distillation

We distill a Pythia-1.4B teacher into a Pythia-160M student on WikiText-103.³ Standard KD combines next-token cross-entropy with teacher matching⁴:

$$\mathcal{L} = \alpha \mathcal{L}_{\mathcal{CE}} + (1 - \alpha) \mathcal{L}_{\mathcal{KD}}.$$

Baselines are Full KD, fixed Top- K KD, and sampling-based sparse KD¹. Our method treats sparsity as teacher-signal filtering, not just compression: it keeps more tokens when the teacher is uncertain and downweights KD when the retained head captures little probability mass, motivated by diffuse teacher tails.²

For each token position t , compute teacher entropy:

$$H_t = - \sum_i p_T(i | x_{<t}) \log p_T(i | x_{<t}),$$

then choose:

$$K_t \in \{4, 8, 16\}.$$

Low-entropy tokens use smaller K , while high-entropy tokens use larger K . Let $\mathcal{V}_{K_t}$ be the retained teacher Top- K_t set. We renormalize the sparse target:

$$\widetilde{p}_T(i | x_{<t}) = \frac{p_T(i | x_{<t})}{\sum_{j \in \mathcal{V}_{K_t}} p_T(j | x_{<t})}, \quad i \in \mathcal{V}_{K_t}.$$

The adaptive KD loss is then:

$$\mathcal{L}_{adapt}^{(t)} = \sum_{i \in \mathcal{V}_{K_t}} \widetilde{p}_T(i | x_{<t}) \log \frac{\widetilde{p}_T(i | x_{<t})}{p_S(i | x_{<t})}.$$

(II) Adaptive Top- K Distillation with Head-Mass-Weighting

Adaptive Top- K decides how many teacher tokens to keep, but after renormalization, every sparse target sums to 1 even if it captures very different amounts of the original teacher distribution. We therefore use the retained head mass as a token-level confidence score:

$$m_t = \sum_{i \in \mathcal{V}_{K_t}} p_T(i | x_{<t}).$$

When m_t is high, the retained Top- K support captures most of the teacher distribution, so the sparse target is a reliable KD signal. When m_t is low, much of the teacher mass lies in the omitted tail, so the renormalized sparse target may overstate the teacher’s confidence. We downweight such positions using

$$w_t = \max(m_t, w_{min}),$$

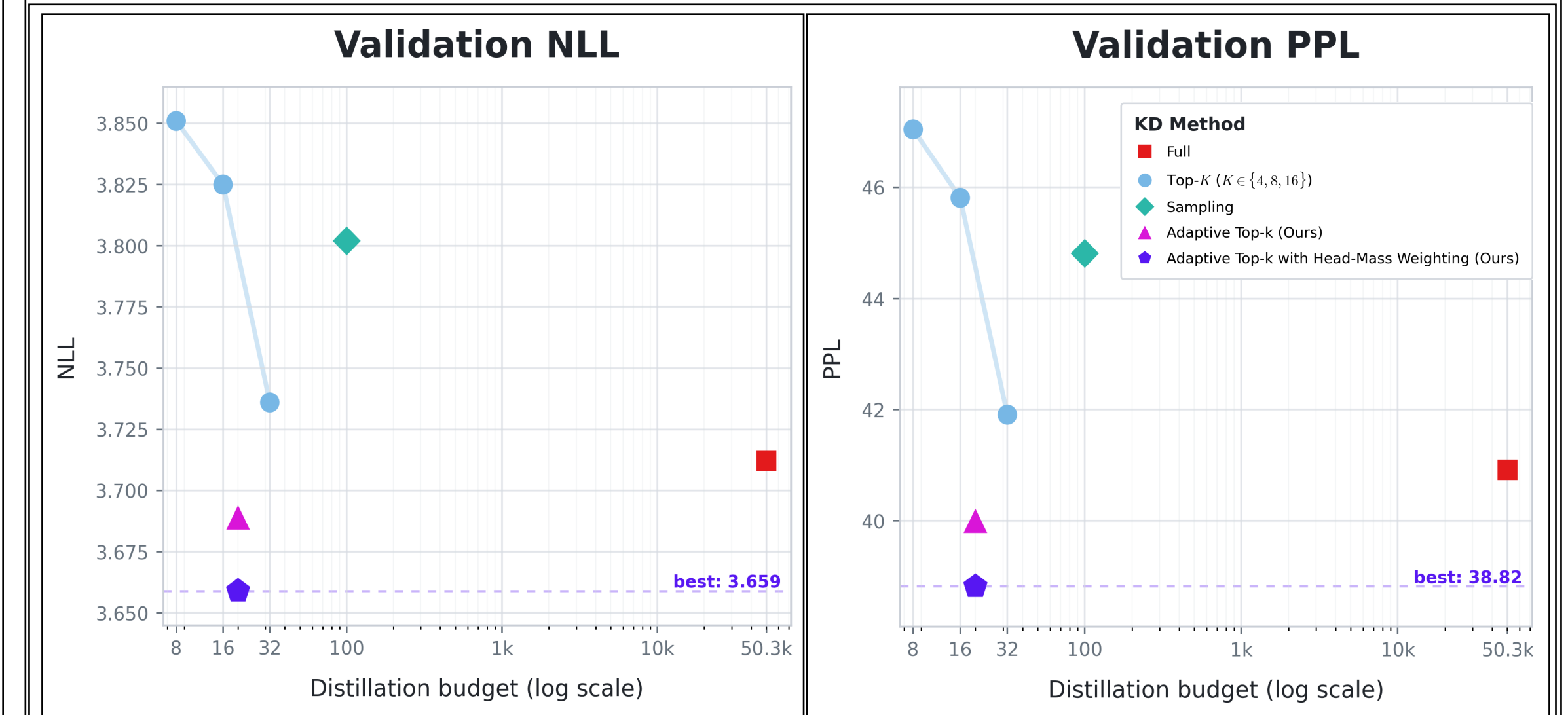
and train with

$$\mathcal{L}_{wkd} = \frac{1}{T} \sum_{t=1}^T w_t \mathcal{L}_{adapt}^{(t)}.$$

Thus, the final training objective becomes:

$$\mathcal{L} = \alpha \mathcal{L}_{\mathcal{CE}} + (1 - \alpha) \mathcal{L}_{wkd}.$$

RESULTS



Method	Budget	Val. NLL ↓	Val. PPL ↓
<i>Reference</i>			
Raw Student	—	4.252	70.22
Teacher	—	2.911	18.37
<i>Baselines</i>			
Full KD	50,277	3.712	40.92
Top-4 KD	8	3.851	47.04
Top-8 KD	16	3.825	45.81
Top-16 KD	32	3.736	41.91
Sampling KD	100	3.802	44.81
<i>Ours</i>			
Adapt. Top- K	20	3.689	40.00
+ Head-Mass Wt.	20	3.659	38.82

Table 1: Main quantitative results on WikiText-103. Budget denotes the approximate number of stored teacher-derived scalars per token. All baselines are reported using their best-performing configuration within the same tuning budget. Adaptive methods use variable token-wise budgets, so we report the average stored values per token. Lower is better.

CONCLUSIONS

- Full-distribution KD may not always be the strongest supervision signal for a capacity-limited student.
- Our adaptive sparse methods outperformed vanilla Top- K , sampling, and full KD baselines in our setup.
- These results were achieved using about 2,514× fewer stored teacher values than Full KD.
- The advantage may stem from filtering the teacher’s diffuse tail distribution, yielding a sharper training signal.
- Future work should replace our heuristic entropy buckets with learned token-wise budget allocation and explore richer confidence-weighting strategies beyond retained head mass.
- More broadly, sparse KD methods for small students may benefit from optimizing not only *how little* teacher information can be stored, but *how salient* those parts of the teacher distribution can be for the student.